

Comparative Analysis of Spectral Approaches to Feature Extraction for EEG-Based Motor Imagery Classification

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Abstract—The quantification of the spectral content of electroencephalogram (EEG) recordings has a substantial role in clinical and scientific applications. It is of particular relevance in the analysis of event-related brain oscillatory responses. This work is focused on the identification and quantification of relevant frequency patterns in motor imagery (MI) related EEGs utilized for brain–computer interface (BCI) purposes. The main objective of the paper is to perform comparative analysis of different approaches to spectral signal representation such as power spectral density (PSD) techniques, atomic decompositions, time-frequency (t-f) energy distributions, continuous and discrete wavelet approaches, from which band power features can be extracted and used in the framework of MI classification. The emphasis is on identifying discriminative properties of the feature sets representing EEG trials recorded during imagination of either left- or right-hand movement. Feature separability is quantified in the offline study using the classification accuracy (CA) rate obtained with linear and nonlinear classifiers. PSD approaches demonstrate the most consistent robustness and effectiveness in extracting the distinctive spectral patterns for accurately discriminating between left and right MI induced EEGs. This observation is based on an analysis of data recorded from eleven subjects over two sessions of BCI experiments. In addition, generalization capabilities of the classifiers reflected in their intersession performance are discussed in the paper.

Index Terms—Alternative communication, brain–computer interface (BCI), electroencephalogram (EEG), spectral analysis, time-frequency (t-f) analysis, wavelet transforms.

I. INTRODUCTION

THE electroencephalogram (EEG) is one of the most clinically and scientifically exploited signals recorded from humans. Hence, its quantification plays a prominent role in brain studies [1]. In particular, the examination of the frequency content of EEG signals has been recognized as the most preponderant approach to the problem of extracting knowledge of the brain dynamics [1], [2] and thus it has been heavily exploited in the study of the brain phenomena related to motor imagery (MI) [3]. The outcome of this study is crucial for successful development of brain–computer interface (BCI) technology, which is

aimed at replacing the impaired functionality of a human neuromuscular system with computer-based analysis of the EEG and establishing a communication channel independent of peripheral nerves and muscles [4]. Most BCIs utilize EEG features that are well-defined physiologically, which include oscillations in neuronal circuits or potentials evoked by particular stimuli [4]. In this work, lateralized changes in the signal's spectral content within μ (8–12 Hz) and β (18–25 Hz) sensorimotor rhythms due to imagination of left- or right-hand movement are exploited for MI discrimination. The underlying brain phenomena is referred to as event-related de-synchronization (ERD) and event-related synchronization (ERS) [1], [3].

The task of EEG spectral quantification is particularly challenging considering the complexity of the dynamics of nonstationary EEG [1]. It is required that the time variation of the relevant frequency components is accounted for. Statistical spectrum estimation methods, both parametric [5] and nonparametric [6], [7], have been extensively utilized for EEG analysis in studies of MIs. Digital filtering, commonly applied in BCIs, e.g., [8], represents a different concept of feature extraction. Instead of estimating spectral content of signal segments, digital filters select their rhythmical components depending on the frequency band of interest. Joint time-frequency (t-f) and time-scale (t-s) techniques are considered to be particularly well suited for analysis of EEGs as representative examples of signals with time-varying spectral content [2]. Hence, quadratic t-f distributions and wavelet-based methods have also been exploited in BCI, e.g., [9], [10].

EEG feature extraction approaches are clearly dominated by methods estimating the signal's energy distributed in the frequency, t-f or t-s domains. However, to the best of authors' knowledge, a comparative analysis of a sizeable set of these methods applied to the same data in the BCI framework using the state-of-the-art linear and nonlinear BCI classifiers has not been undertaken yet. Most comparative evaluations reported in the literature are confined to a small number of techniques or only one classifier [8], [11], [12]. Alternatively, studies involving an examination of different feature types are concerned with the problem of their selection from multichannel EEG recordings, which does not provide much insight into the relation between the nature of the given t-f signal representation and discriminative properties of the features generated [7], [13]. Therefore, this paper is aimed at a more systematic analysis of different approaches to quantifying the frequency content of the EEG trials recorded for BCI purposes. They are categorized into four groups:

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spectral estimation methods, atomic decompositions, quadratic t-f distributions, and wavelet-based techniques. Wavelets are considered in this work despite the fact that they map a signal into the t-s domain [2] since the scale can be interpreted and related to the frequency domain. All the techniques are applied within a uniform feature extraction procedure to quantify the signal energy in the frequency bands of interest. For comparative purposes, autoregressive (AR) based feature extraction is employed, where the AR coefficients are used as features of the EEG signal. AR modeling approach has attracted a lot of interest in BCI (e.g., [12]) since it does not involve meticulous tuning of subject-specific frequency bands. Both linear and nonlinear approaches to the MI classification task are adopted to investigate the separability of the features. Linear discriminant analysis (LDA) [14], regularized Fischer discriminant (RFD) [15], and support vector machines (SVMs) [16] with linear and nonlinear kernels are employed as classifiers. In this offline study, classification is carried out discretely at the end of every signal trial, as in [9] and [17], and similar to the delayed mode in [18]. Classification accuracy (CA) serves as the primary performance measure.

The major outcome of the comparative analysis reported in this work is the observation that power spectral density (PSD) approaches are the most robust and consistent in extracting the distinctive MI-related EEG spectral patterns over two BCI sessions for eleven subjects. Moreover, on average, the linear classifiers have shown superior generalization capability to that of the SVMs with Gaussian kernels.

The paper is organized as follows. Section II outlines the methodology used with emphasis on the techniques applied to extract spectral information from the EEG. The results of the comparative analysis of different combinations of the EEG features and the BCI classifiers are demonstrated in Section III and discussed in Section IV. Conclusions and scope of future work are presented in Section V.

II. METHODS

A. Data Acquisition

There are two sets of EEG data utilized in this work. The first dataset was obtained from the Institute of Human-Computer Interfaces, Graz University of Technology. The EEG signals were recorded from three subjects (S1, S2, and S3) in a timed experimental recording procedure where the subjects were instructed to imagine moving the left and the right hand in accordance with a directional cue displayed on a computer monitor [Fig. 1(a)]. Each trial was 8 s in length. A fixation cross was displayed from $t = 0$ s to $t = 3$ s. The beginning of a trial was marked by acoustic stimulus at $t = 2$ s. Next, an arrow (left or right) was displayed as a cue at $t = 3$ s. Therefore, the segment of the data recorded after $t = 3$ s of each trial is considered as event related and is used for offline analysis. The recordings were made with a g.tec amplifier and AgCl electrodes over two consecutive sessions, each session consisting of 140 trials for S1 and 160 trials for S2 and S3 with equal number of trials representing two classes [19]. Two bipolar EEG channels were measured over C3 and C4 locations (two electrodes placed 2.5 cm anterior and posterior to positions C3 and C4) according to the international standard 10/20 system [1]. The EEGs were then

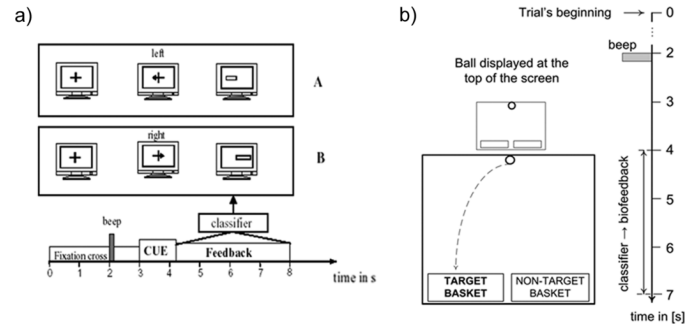


Fig. 1. Data recording in a) Graz BCI paradigm [12] and b) BCI basket paradigm.

sampled at a frequency (f_s) of 128 Hz and band-pass filtered in the range 0.5–30 Hz. The experimental settings for recording the data under consideration are described in greater detail in [19].

The second EEG dataset was acquired at the Intelligent Systems Research Centre, University of Ulster at Magee using the same g.tec equipment as that used by the Graz BCI group. The EEG data were obtained from eight subjects ($S_I - S_{VIII}$) over ten (for subjects $S_I - S_{VI}$) or fourteen (for subjects S_{VII} and S_{VIII}) 160-trial (balanced) sessions. Two consecutive sessions for each subject were arbitrarily selected for extensive offline analysis reported in this paper. Trials were sampled at $f_s = 125$ Hz and those with distinctive muscle or eye artifacts were removed by trained staff. The subjects were asked to imagine moving the left- and the right-hand depending on the horizontal location (left/right) of a target basket displayed at the bottom of a monitor screen [Fig. 1(b)]. Each trial was 7 s in length. A ball was displayed at the top of the screen from $t = 0$ s to $t = 3$ s. In the meantime, at $t = 2$ s acoustic stimulus signified the beginning of a trial and then the baskets (target in green and nontarget in red) were displayed at $t = 3$ s. At $t = 4$ s the ball started moving to the bottom of the screen. The segment of the data recorded after $t = 3$ s of each trial was analyzed in this work. The horizontal component of the ball movement was continuously controlled in online experiments by the subject via the biofeedback mechanism. However, as mentioned, the comparative EEG analysis discussed in this paper was carried out offline with classification performed at the end of every trial. This approach with delayed classification until the entire trial is processed resembles the concept of single MI-related EEG trial discrimination [9], [17] as opposed to continuous classification adopted in online experiments.

The EEG datasets obtained from two different BCI laboratories, the Graz and ISRC BCI, were evaluated in the same framework in this work as they share the same characteristics accounted for in offline analysis. In addition, since the Graz recordings were acquired from subjects with more experience and longer history of participation in BCI experiments when compared to the participants of the recent study conducted in the ISRC BCI laboratory, this mixture of EEG datasets allowed for capturing general trends in the performance of spectral EEG feature extraction methods and BCI classifiers.

B. Feature Extraction Procedure

The frequency bands in which ERS and ERD occur providing discriminative information for classification vary from subject

to subject [3], [8]. The resultant characteristic changes in the EEG spectral content around μ (8–12 Hz) and central β (18–25 Hz) ranges depend on the associated neural structure and the locations where the neurons are active during MI. The experiments showed that for the eleven subjects examined, there was ERD of the μ rhythm on the contralateral side and a slight ERS in the central β rhythm on the ipsilateral hemisphere. This hemispheric asymmetry reflected in the EEGs is exploited to differentiate between the MIs.

The quality of the spectral representations examined in this paper is assessed in terms of distinctive properties of the features obtained from windowed EEG signals. To this end, the EEG trials of length N were divided into windows along the time axis, depending on the settings of two parameters: window size, $win.len$, and the amount of overlap, ovl . The number of windows n_{win} could be calculated, as shown below

$$n_{win} = \left\lfloor \frac{N - ovl}{win.len - ovl} \right\rfloor. \quad (1)$$

Next, relevant portions of information corresponding to spectral correlates of ERD/ERS phenomena were extracted separately from each EEG window within a trial using the techniques examined in this paper. The frequency components related to ERD and ERS were merged together to reduce the dimensionality of the feature space (as explained in Sections II-B1–II-B5). Closer analysis performed at initial stages using separate correlates of ERD and ERS revealed instability and variability of ERS patterns in β frequency bands. Therefore, using each feature as a separate dimension of the feature space was abandoned. As a result, a feature vector $\mathbf{r}$ representing a BCI trial was composed of $2 \cdot n_{win}$ elements r_i^j ($i = 1, \dots, n_{win}$) given two recording channels, $j \in \{C3, C4\}$

$$\mathbf{r} = (r_1^{C3}, r_2^{C3}, \dots, r_{n_{win}}^{C3}, r_1^{C4}, r_2^{C4}, \dots, r_{n_{win}}^{C4}). \quad (2)$$

The most reactive frequency bands from which to extract features for the given subjects were tuned by optimizing the CA evaluated with an LDA classifier in a five-fold cross-validation (CV) procedure.

1) *Spectral Estimation Methods*: Spectral density methods extract information from a signal as a stochastic process to describe the distribution of its power in the frequency domain. The PSD is defined as the Fourier transform (FT) of the signal's autocorrelation function provided that the signal is stationary in a wide sense [20]. In practice, a comprehensive statistical characteristic of the random signal is not available and its spectral content can only be estimated from a sequence of time samples. Parametric approaches work on the assumption that the underlying process can be described parametrically as an output of a linear system driven by white noise. In this work, the coefficients of AR model were estimated from segments of EEG time series using the Yule–Walker algorithm [20] and the corresponding PSD was then derived from its frequency response. This parametric spectral estimation technique is referred to throughout the paper as Yule_{PSD}.

The two other methods allowed for nonparametric estimation of the signal's PSD from its periodogram [20]. It was either calculated as the FT of the windowed autocorrelation

function of the signal ($period_{PSD}$) or using Welch's approach (Welch_{PSD}), referred to as the modified periodogram [20], consisting in averaging multiple spectral estimates for shorter data segments and windowing. The average power in a given frequency band was obtained from a rectangle approximation of the integral of the signal's PSD. This led to the calculation of r_i^j in (2).

2) *Atomic Decompositions*: A t-f representation produced by methods falling into this category has the form of a linear composition of elementary components referred to as atoms. In this work, two techniques were applied, the discrete short-time FT (STFT) and the S transform. The major limitation of the STFT is a trade-off between time and frequency resolutions depending on the size T_{win} of the window along the time axis [1]. Here, a Gaussian window was applied. Since it is kept constant throughout the signal analysis, the balance between time and frequency localization capabilities should be determined beforehand.

The S transform combines the separate strengths of the STFT and wavelet transform [21]. First, the signal is decomposed into t-f atoms and second, since the analysis window changes with frequency it can provide satisfactory resolution in both domains of interest allowing for better time localization at higher frequency components. The discrete formula of the S transform is the following:

$$S_x[n, m] = \frac{1}{N} \sum_{k=0}^{N-1} x[k] h \left[n - k, \frac{m}{N} \right] \exp \left(-j2\pi k \frac{m}{N} \right), \quad (3)$$

where $x[k]$ is a sampled signal $x(t)$ (with N samples) and $h[n, m]$ is a discrete form of the window $h(t, f)$. It is analogous to that of the STFT with the difference in the form of the Gaussian window h , which is additionally parameterized with the oscillations frequency f

$$h(t, f) = \exp \left(-\frac{(tf)^2}{2} \right). \quad (4)$$

The concepts of wavelet and STFT analysis in S transform are incorporated by separating the kernel into two parts: the slowly varying frame h and the oscillatory exponential element which chooses frequency bins being localized as in the STFT [21]. Unlike the frame h , the oscillatory component is not translated over the time axis, which allows for independent localization of the phase and the amplitude spectrum. The implementation of the decomposition methods was done using algorithms based on the fast FT (FFT) [2].

In order to determine the feature vector components, r_i^j s in (2), the amplitudes of the atoms representing an EEG window were time averaged over its length, $win.len$, and then the square norm of these mean values F within the frequency bands of interest (adjusted μ and β) was calculated (cf. Fig. 2).

3) *Quadratic Energy Distributions*: Unlike the linear atomic decomposition that deals with elementary t-f components, the energy-based approach consists in distributing the energy of the signal in the joint t-f domain. The Wigner–Ville distribution (WVD) served as a starting point in this research. It belongs, like other transforms studied here, to the Cohen's class of the quadratic t-f representations based on second-order moments of the signal, covariant by shifts in time and frequency [2], [22].

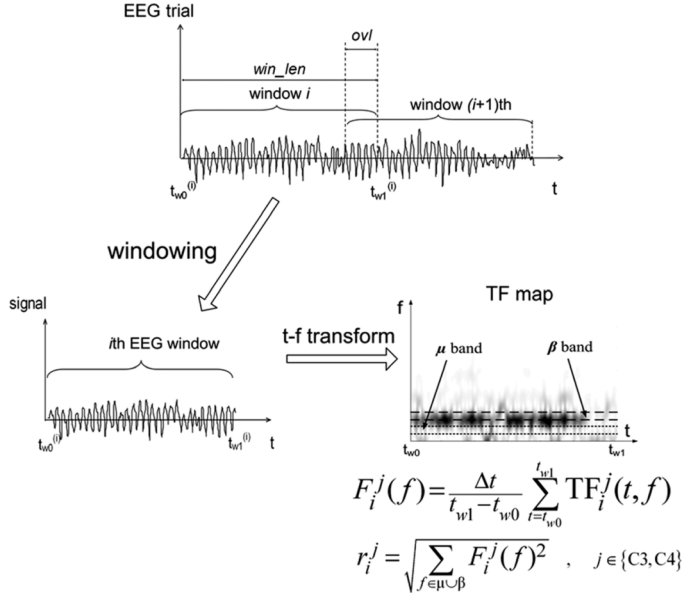


Fig. 2. Schematic presentation of the EEG feature extraction procedure with atomic decomposition techniques and quadratic energy distribution approaches.

For a discrete time analytic signal $x[k]$ (using the same symbol nomenclature as in (3) with $x^*[k]$ being the conjugate signal), the WVD transform is theoretically defined as [23]

$$WVD_x[n, m] = \sum_{k=-N/2}^{N/2} x[n+k]x^*[n-k] \exp\left(-j4\pi k \frac{m}{N}\right). \quad (5)$$

As a bilinear transform, the WVD introduces cross terms between energy components in different t-f regions according to the quadratic superposition principle [2]. The resultant interference terms may overlap with the desirable auto-terms thus likely leading to misinterpretation. This can be alleviated by the windowing operation over the time axis. It is equivalent to frequency smoothing of the WVD (with a parameter F_{win} as the window length) and results in the pseudo WVD (PWVD) [22]. In consequence, at the cost of lower t-f resolution, the interferences are attenuated. More flexibility in the smoothing can be gained by separable smoothing in time and frequency (with T_{win} and F_{win} as the length of the respective windows), which leads to the smoothed-pseudo WVD (SPWVD) [22]. It facilitates reaching the compromise between the decrease in the t-f resolution and clearer readability of the energy map due to the attenuation of the interferences.

Another group of the quadratic t-f representations examined in this paper are reduced interference distributions (RIDs). They apply cross terms filtering to the WVD in the so-called ambiguity plane [2]. In particular, RIDs as members of the Cohen's class can be represented using the 2-D FT of the WVD, $A_x(\xi, \tau)$, and the parameterization function $\varphi(\xi, \tau)$ in the following continuous form:

$$RID_x(t, f; \varphi) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \varphi(\xi, \tau) A_x(\xi, \tau) \times \exp(-j2\pi(f\tau + \xi t)) d\tau d\xi. \quad (6)$$

Formulation of A_x in the space with new relative coordinates, Doppler ξ and delay τ (corresponding to the f and t dimensions) facilitates representing the original signal in the ambiguity plane with the cross terms shifted towards the origin. This allows for interpretation of the function φ in (6) as a low-pass filter that attempts to reject the cross terms and leave the signal terms unchanged. The shape of the filter can be adapted to the analysis of the desirable signal components depending on the nature of the signal. For RIDs, $\varphi(\xi, \tau) = \Phi(\xi\tau)$ where Φ is a monotonously decreasing function such that $\Phi(0) = 1$ to ensure that the parameterization function is a low-pass filter in the ambiguity plane. Depending on the canonical form of Φ , the resultant RIDs have different cross term rejection capabilities. In this work, three RID distributions, Choi-Williams (CW_{RID}), Born-Jordan (BJ_{RID}), and Zhao-Atlas-Marks (ZAM_{RID}), were examined due to their successful applications in the EEG analysis [2]. Similar to the SPWVD, the filters for CW_{RID} , BJ_{RID} , and ZAM_{RID} were parameterized with the length of the time and frequency smoothing windows T_{win} and F_{win} .

The EEG feature vector components, r_i^j s in (2), were obtained carrying out analogous operations to the procedure used for atomic decompositions (cf. Fig. 2). In consequence, the square norm of the time-averaged energy components within the frequency band of interest served as r_i^j . The quadratic transformations were performed on segments of an EEG trial using the implementation available in [22].

4) *Wavelet-Based Approaches:* Wavelet-based methods have been heavily exploited in EEG signal analysis (for a review, refer to [2]) and, consequently, in BCI research, e.g., [9], [10], to capture the spectral dynamics of EEG trials. Their computational speed and capability to discriminate both the temporal and spectral domain features of signals are important assets [1]. The concept of a multiscale approximation associated with the wavelet transform (WT) allows for effective localization of signal components with various spectro-temporal characteristics. In this light, signal decomposition into a linear combination of shifted and scaled versions of the mother wavelet can be considered as its harmonic analysis with multiband filters of varying widths. Hence, wavelet-based approaches do not suffer the t-f resolution trade-off inherent to the STFT and thus more naturally lend themselves to analysis of nonstationary time series like EEGs. In this work, two WT-related methods were utilized: a generalized form of discrete wavelet decomposition, referred to as a wavelet packet (WP) transform (WPT), and the discrete-time form of the continuous CWT.

WPT offers flexibility in decomposing the signal into frequency bands of various sizes and concise representation within a dyadic set of the wavelet scales and translations. It is particularly important in the analysis of EEGs for BCI purposes since it facilitates suitable adjustments to the reactive ERD/ERS bands for individual subjects. This need dictated the basis selection for the WP tree in this work. Appropriate WP coefficients at scale levels corresponding to the frequency bands of interest were squared and summed to represent the energy of the decomposed segment of an EEG trial in the specific ERD/ERS bands. The sum served then as the EEG feature vector component r_i^j . Symlet (sym) and Daubechies (db) mother wavelets of different orders were used.

TABLE I
LIST OF SPECTRAL TECHNIQUES USED IN FEATURE EXTRACTION

Technique	Acronym	Parameters
Yule-Walker PSD est.	Yule _{PSD}	n_{AR}
Periodogram PSD est.	Period _{PSD}	—
Welch periodogram PSD est.	Welch _{PSD}	—
Short-time Fourier transform	STFT	T_{win}
S transform	S _{TRANS}	—
Wigner-Ville distr.	WVD	—
Pseudo Wigner-Ville distr.	PWVD	F_{win}
Smoothed pseudo Wigner-Ville distr.	SPWVD	F_{win}, T_{win}
Choi-Williams RID	CW _{RID}	F_{win}, T_{win}
Born-Jordan RID	BJ _{RID}	F_{win}, T_{win}
Zhao-Atlas-Marks RID	ZAM _{RID}	F_{win}, T_{win}
Continuous Morlet wavelet transform	Morlet _{CWT}	—
Wavelet packets with sym wavelets	Sym _{WP}	n_{DWT}
Wavelet packets with db wavelets	Db _{WP}	n_{DWT}
Autoregressive model	Yule _{AR}	n_{AR}

In the second WT-related approach adopted in this work, the redundancy of the CWT was exploited to enhance localization of the relevant spectral components of the signal. The discrete-time complex Morlet wavelets with the scales finely adjusted to the individualized ERD/ERS bands were applied to EEG segments. This was performed using the relation between the scale, the main receptive frequency, the effective frequency width and the characteristic eigenfrequency of the wavelet [10]. As a result, every time point within a signal segment was associated with two CWT coefficients, W_μ and W_β , corresponding to the desirable frequency bands within μ and β . They were squared and time averaged over the window length in each band. The square norm of the two resultant quantities, $\overline{W}_\mu^2$ and $\overline{W}_\beta^2$, was used then as the EEG feature vector component, $r_i^j = \sqrt{\overline{W}_\mu^2 + \overline{W}_\beta^2}$.

5) *AR Model*: The AR parameterization of a signal, which gives rise to a maximum entropy spectral estimate, has been heavily exploited in EEG modeling (for a review, refer to [1]) and applied in BCI research to date, e.g., [11]. In consequence, the approach consisting in representing quasi-stationary segments of the EEG trials by a set of AR coefficients was employed in this work for comparative purposes. From a number of available estimation methods (for a review, refer to [24]) the Yule-Walker algorithm [20] was chosen to determine parameters of an AR model of a given order n_{AR} . An EEG feature component corresponding to an EEG trial segment has the form of the subvector $\mathbf{r}_i^j$ consisting of n_{AR} coefficients $a^{i,j}$

$$\mathbf{r}_i^j = (a_1^{i,j}, a_2^{i,j}, \dots, a_{n_{AR}}^{i,j}), \quad i \in \{1, \dots, n_{win}\}, j \in \{C3, C4\}. \quad (7)$$

All the spectral techniques discussed above are listed in Table I with their acronyms and sets of adjustable parameters.

C. MI Classification

The aim of BCI classification is to assign EEG trials to the classes of the associated mental tasks. The separability of the EEG features extracted with the techniques being compared was analysed using LDA, RFD, and SVMs with linear and nonlinear

TABLE II
LIST OF THE CLASSIFIERS APPLIED IN THE STUDY

	Classifier	Parameters	Acronyms
Linear	LDA	—	LDA
	RFD	C	RFD
	SVM with linear kernel (SVM _{lin})	C	L1-SVM _{lin} L2-SVM _{lin} LS-SVM _{lin}
	SVM with inhomogeneous Gaussian kernel (SVM _{lin})	$C, \sigma_1, \dots, \sigma_L$	L1-SVM _{GN} L2-SVM _{GN} LS-SVM _{GN}
Nonlinear	SVM with homogeneous Gaussian kernel ($\sigma_1 = \dots = \sigma_L = \sigma$) (SVM _{Gauss})	C, σ	L1-SVM _G L2-SVM _G LS-SVM _G

kernels, as listed in Table II. The CA rate was used as a quantifier of the distinctive properties of the features.

LDA is a Bayes optimal classifier provided that the distributions of features in each of two classes are normal with the same covariance matrix [14]. This assumption was fulfilled for only one feature set (S1). A linear separating hyperplane was determined as a result of maximizing the ratio of the interclass variance to the intraclass variance. To enhance robustness of this linear classifier, its regularized version, RFD, was also applied. Regularization facilitated the reduction of the disadvantageous effects of outliers and strong noise on the generalization ability of the classifier [15]. Since RFD is formulated as a quadratic programming (QP) problem [15], it was implemented using the Matlab QP solver. Unlike an LDA classifier, RFD requires one regularization parameter, which was selected using a five-fold CV procedure [25].

SVMs have also been heavily used in BCI research [11], [26]. This machine learning approach based on the idea of structural risk minimization [27] has proven to perform well in binary classification tasks offering exceptional generalization capability [16], which is ensured as a result of searching for a separating hyperplane with the largest margin between classes [27]. In the so-called soft version of an SVM [16], costs of misclassification of a portion of data points in a certain neighborhood of the decision boundary are neglected. The size of the neighborhood is controlled by a corresponding regularization parameter C that determines the trade-off between the training error and the size of the margin between classes. Its selection is crucial for the effective setup of an SVM classifier. The separating hyperplane is found by solving a QP problem with one global optimum [16]. Depending on the formulation of the optimization problem, different SVM variants can be adopted. In this work, so-called L1-SVM and L2-SVM were investigated first. They refer to SVMs with linear (ℓ_1 -norm) and quadratic (ℓ_2 -norm) penalization of misclassified examples outside the given neighborhood [16]. They were implemented using the SVM-KM toolbox [28]. For comparison, a least squares SVM (LS-SVM) was also applied here using the implementation in [29]. It is a reformulation of standard SVMs that leads to a less computationally demanding set of linear equations than a QP problem [29].

One of the strengths of SVMs exploited in this work is the fact that they are kernel machines, thus allowing for straightforward transformation into nonlinear classifiers [16]. A nonlinear kernel function is implicitly utilized to map the original input space to a higher dimensional feature space in which the problem is more likely to be linearly separable. Two nonlinear kernels were employed—Gaussian and polynomial [16]. However, due to the poor performance of SVMs with the polynomial mapping when applied to the MI classification, only the following Gaussian kernel function $k(\mathbf{x}, \mathbf{y})$ was used:

$$k(\mathbf{x}, \mathbf{y}) = \exp \left(\sum_{i=1}^L \left(\frac{x_i - y_i}{\sigma_i} \right)^2 \right). \quad (8)$$

It is emphasized that the kernel is inhomogeneous if the components of the L -dimensional vectors, $\mathbf{x} = (x_1, \dots, x_L)$ and $\mathbf{y} = (y_1, \dots, y_L)$, have different scaling factors σ_i ($i = 1, \dots, L$).

Selecting an optimal parameter set poses a considerable difficulty but is crucial for effective utilization of SVMs. An SVM classifier is controlled by multiple parameters such as the regularization constant C and kernel parameters σ_i s, which need to be carefully chosen to minimize the generalization error [27]. In this work, it was estimated using a CV scheme and a theoretical radius-margin bound [30]. The search for an optimal set of the parameters was performed with a genetic algorithm (GA) due to its ability to efficiently solve optimization problems given vast, multidimensional and highly irregular search spaces [31]. A roulette wheel selector along with a two-point crossover (with probability rate $P_c = 0.8$) and binary mutation ($P_m = 0.12$) operators were utilized. Through a smaller-scale pilot study employing micro-GAs, this setup was found to deliver satisfactory performance in terms of computational time and the quality of the resultant SVM classifier. Moreover, an elitist strategy was applied with a replacement rate of $R = 0.1$ (for the population size of $N_{\text{pop}} = 50$). The GA search was halted after $N_{\text{gen}} = 30$ generations due to a considerable drop in the convergence rate. Two variants of the objective function to be minimized were used. The first was based solely on the average rate of the classification error evaluated using multiple-run (here, 10) five-fold CV whereas the second was defined as a weighted sum of this CV error estimate and the theoretical upper bound on the generalization error referred to as a radius-margin bound. The outcome of their application is further discussed in Section IV. An extensive examination of the stochastic nature of the GA optimization procedure facilitated the observation that ten randomly initiated runs of the algorithm resulted in low variance average values of both objective functions applied in the experiments. Moreover, after ten runs of the GA search scheme it became clear that the regions of the solution space explored by the algorithm at final stages of the optimization process started to overlap to a considerable degree. Therefore, GA-based algorithm was run ten times to identify the desirable parameter setup of an SVM classifier for each given feature set. In addition, gradient-based minimization of the estimate of the generalization error bound [30] was conducted for the resultant SVM designs following GA search to enhance local optimization capabilities of the algorithm. It did not lead however to finding better solutions in terms of the CV estimate of the generalization error

than those originally selected by the GAs, which confirmed the desired optimality of the GA solutions in the given regions of the SVM parameter space.

III. RESULTS

The process of parameter selection for each of the feature extraction methods was conducted first. The most reactive frequency bands, a setup of the windowing procedure (*winLen*, *ovl*) and a set of adjustable parameters of the spectral techniques listed in Table I were optimized using a ten-run five-fold CV procedure on Session I data with the average CA as a performance measure. A preliminary study showed consistent trends of these parameter setups across different classifiers, which allowed for further tuning to maximize the interclass feature separability quantified using LDA only.

Next, the overall separability of the EEG features was evaluated on Session I data in terms of the discrimination error estimate made using the inner–outer CV scheme (five-fold splits with multiple runs) with all the classifiers under consideration. To this end, Session I data was divided into five outer folds and the classifiers' parameters were optimized in the GA framework only on the training set, inner fold (again five-fold inner CV split), of each outer split. In addition, both the outer and inner five-fold splits were performed in multiple runs after random data shuffling. The final overall CA rate was obtained as the outer-test-fold average across multiple runs.

Analogously to the selection procedure used for spectral parameters, another ten-run five-fold CV procedure was utilized (without inner–outer fold splits) within the framework of the GA to identify the parameter setup of the SVM/RFD classifiers for use in the second experiment—a single-pass test on Session II data. This experiment facilitated verification of the session-to-session generalization performance. It is crucial for BCI systems since the dynamics of EEG recordings acquired at distant times (different sessions) is known to vary considerably. These variations degrade the intersession performance of BCIs [32]. A key question is to what extent the spectral methods applied in this work capture the most salient and persistent (session-to-session) characteristics of EEGs in the context of MI classification. Generalization capabilities of the classifiers play an important role in this regard also. The test of session-to-session performance was conducted using the unseen Session II data. To this end, optimal configurations of the parameters identified using CV on Session I data were kept unchanged. The classifiers were then trained on the features extracted from Session I and tested on those from Session II.

The results of this multisubject study evaluated using the inner–outer CV over Session I and the single-pass test on Session II are presented in Fig. 3(a) and (b), respectively. The CA rates were averaged over eleven subjects for every technique and classifier type. For SVMs, the classifiers performing best on Session I in each category, linear and nonlinear, were considered as SVM_{lin} and $\text{SVM}_{\text{Gauss}}$. It was noted that the Session I performance estimated with the inner–outer CV scheme [Fig. 3(a)] showed similar trends to the mean CAs obtained from the ordinary ten-run five-fold CV during the selection of the classifier's parameters (not reported here).

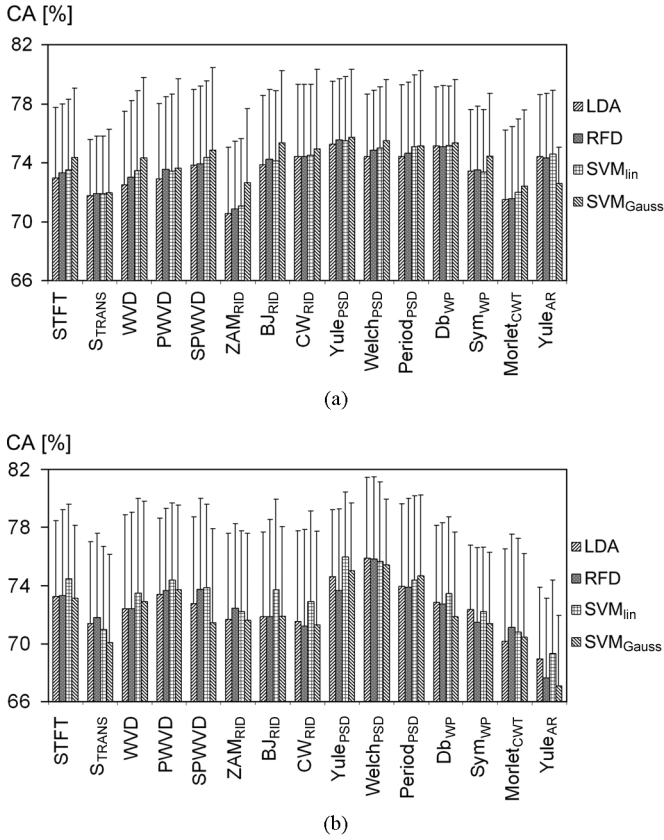


Fig. 3. (a) Mean CA rates obtained on Session I data using the inner-outer CV scheme (five-fold splits with multiple runs) and averaged over 11 subjects. Vertical lines denote the intersubject standard deviations of the respective mean CA values. (b) Average CA rates obtained in the single-pass test on Session II data for 11 subjects. Vertical lines denote the intersubject standard deviations of the respective mean CA values.

Fig. 4(a) and (b) depicts the average classification performance associated with different categories of frequency techniques: atomic decompositions, WV energy distributions, RIDs, PSD estimators, t-s approaches, and AR modeling. The resultant mean value for each category shown in the form of a bar was calculated for every subject as the average CA rate obtained with all the classifiers and the category-specific features. Again, the mean and the standard deviation across the subjects are shown for Session I and Session II. In addition, Table III lists the best and average CA rates obtained for each subject across all the classifiers. It illustrates intersubject differences in performance on Session I and Session II data.

A comparative analysis of the performance of the classifiers employed in this work, including all types of SVMs listed in Table II, was also conducted. The corresponding CAs were averaged over the spectral methods and their means obtained for eleven subjects are presented along with the standard deviations in Fig. 5 for Session I and Session II data.

IV. DISCUSSION

The results of this multisubject study were processed independently for two sessions of the data in the framework of two-way analysis of variance (ANOVA) with repeated measures [33]. This approach facilitated effective accounting for rather high intersubject variance. The two within-subject

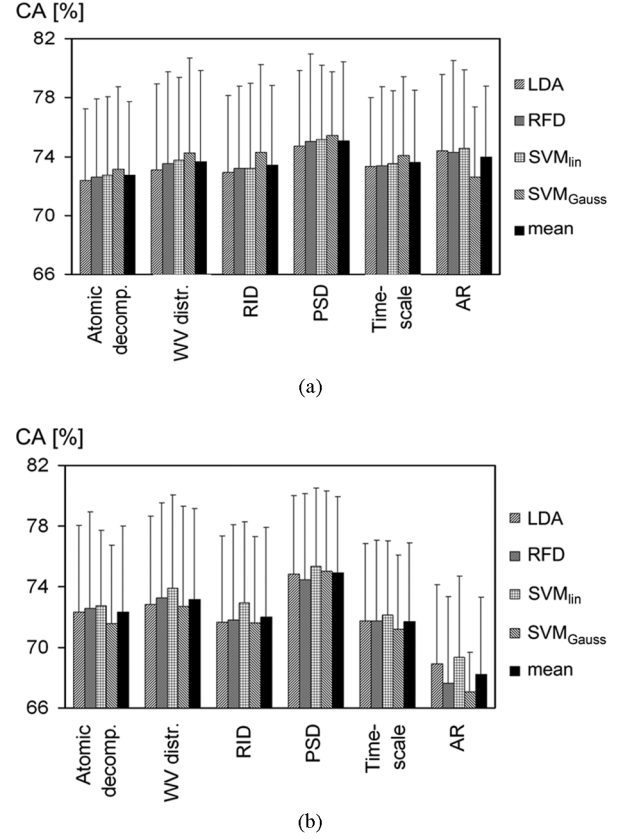


Fig. 4. Comparative analysis of the categories of frequency techniques in terms of the average CA rates obtained over 11 subjects with different classifiers: (a) on session I data (inner-outer CV scheme) and (b) on session II data (single-pass test). Vertical lines denote the intersubject standard deviations of the respective mean CA values.

TABLE III
LIST OF THE AVERAGE AND BEST CA RATES OBTAINED FOR EACH SUBJECT ACROSS ALL THE CLASSIFIERS IN SESSIONS I AND II. OPTIMAL FEATURE TYPES CORRESPONDING TO THE BEST RESULTS ARE PRESENTED IN COLUMNS "FEATURE" FOR BOTH SESSIONS

Subject	Session I – CV			Session II – test		
	Mean CA [%]	Best CA [%]	Feature	Mean CA [%]	Best CA [%]	Feature
S1	88.0	90.5	Sym _{WP}	86.0	88.4	Welch _{PSD}
S2	79.5	83.3	Yule _{PSD}	82.7	90.1	period _{PSD}
S3	77.5	82.0	Yule _{PSD}	66.8	73.4	Welch _{PSD}
S _I	65.3	68.8	Db _{WP}	60.0	63.8	Yule _{PSD}
S _{II}	71.4	73.8	WVD	71.8	75.5	CW _{RID}
S _{III}	78.4	83.0	period _{PSD}	73.7	78.6	Welch _{PSD}
S _{IV}	61.9	65.0	Sym _{WP}	67.7	72.2	Yule _{PSD}
S _V	61.7	65.9	Yule _{PSD}	60.1	63.7	Welch _{PSD}
S _{VI}	68.2	75.7	Yule _{AR}	66.5	75.1	Yule _{PSD}
S _{VII}	76.8	80.8	Sym _{WP}	76.7	79.7	Yule _{PSD}
S _{VIII}	86.7	88.3	SPWVD	94.6	96.3	SPWVD

factors were frequency technique (14 types) and classifier, represented by LDA, RFD, and two collective categories of

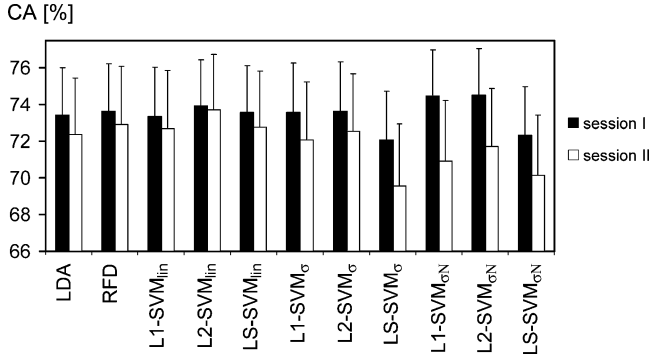


Fig. 5. Comparative analysis of the classifiers' performance averaged over 11 subjects across all the feature extraction methods in each session (inner-outer CV scheme with multiple runs of five-fold splits on Session I and single-pass test on Session II). Vertical lines denote the intersubject standard deviations of the respective mean CA values.

optimal SVM_{Ilin} and optimal SVM_{Gauss}. Yule_{AR} was excluded from this analysis since it introduced an interaction term and violated the assumption of sphericity [33].

Both factors, classifier type and feature type, were found to significantly affect the results of Session I ($p < 0.05$) and Session II ($p < 0.1$) experiments, i.e., the null hypothesis of equality was rejected. No other effects or significant interactions between the factors were detected. Tukey's honestly significant difference criterion [33] was used to perform the factorial post test simultaneous comparison of feature sets and classifiers. In consequence, Yule_{PSD} was found to deliver, on average, significantly higher CA rates than ZAM_{RID} on Session I data (at the significance level of $\alpha = 0.05$). The test results obtained on Session II revealed that two PSD approaches, Welch_{PSD} and Yule_{PSD}, outperformed S_{TRANS} and Morlet_{CWT} ($\alpha = 0.1$). Analysis of the main effect of each classifier demonstrated that the optimal SVM_{Gauss} provided significantly better average performance than LDA ($\alpha = 0.05$) on Session I data. On the other hand, an analogous post-ANOVA multiple comparison of classifiers tested on Session II indicated the inferiority of the optimal SVM_{Gauss} in relation to optimal SVM_{Ilin} ($\alpha = 0.1$).

An analysis of the average classification performance obtained in both sessions with spectral methods grouped in six categories [Fig. 4(a) and (b)] leads to more general conclusions. First, it reveals the overall robustness and consistency of PSD approaches in generating the most distinguishable sets of features from MI induced EEG trials. Moreover, high CA rates produced with the Yule_{AR} approach in the evaluation on Session I data and their radical deterioration in the single-pass test on Session II data deserve special attention. This contrast can be partly explained by excessively high dimensionality ($n_{AR} = 8, \dots, 14$) of the corresponding EEG feature vector and low specificity of distinctive spectral patterns relevant to the intersession classification problem. It has also been observed that the AR feature distribution, which is highly sensitive to noise in a raw signal, markedly varies from one session to the other. Finally, the other four categories of spectral methods displayed comparable performance with consistent trends among classifier types in each session. It is worth noting that S_{TRANS},

ZAM_{RID}, and Morlet_{CWT} produced clearly the least separable features in their respective categories [Fig. 3(a) and (b)].

As can be seen from Table III, optimal spectral feature extraction techniques vary depending on a subject for Session I data whereas when session-to-session performance is considered, PSD approaches, mainly Yule_{PSD} and Welch_{PSD}, clearly dominate. This confirms the overall superior performance of PSD methods with emphasis on their intersession generalization properties.

Fig. 5 facilitates a more detailed comparative analysis of the performance of the classifiers listed in Table II, conducted to complement the outcome of the ANOVA tests. It can be observed that the highest CA rates in the CV evaluation on Session I data were obtained with L1-SVM_{σN} and L2-SVM_{σN} classifiers, most often playing the role of the optimal SVM_{Gauss}. However, their performance in the intersession classification significantly deteriorated and in consequence, SVMs with homogenous kernels, mainly L1-SVM_σ and L2-SVM_σ, were found to be more robust. In the SVM_{Ilin} category, L2-SVM_{Ilin} demonstrated the supreme effectiveness in both experiments and thus in the majority of cases, was considered as the optimal representative. On average, SVM_{Ilin} and RFD proved to be the most reliable approaches in the overall two-session analysis. Although the optimal SVM_{Gauss} yielded the highest CA rates in the CV on Session I, confirmed in the ANOVA framework, its deficient generalization capability in the session-to-session evaluation renders the SVM_{Gauss} approach, particularly with inhomogeneous kernel, less practical in application to MI classification problems.

As mentioned earlier, two objective functions were utilized in the GA-based procedure for optimization of the SVMs' parameters. The combination of the CV error estimate and the theoretical radius margin bound often led to better session-to-session results but at the cost of the performance on Session I. Still, the highest CA rate on Session I data was the criterion for optimal SVM selection among those produced by GA.

The analysis reveals a rather high variance in the CA rates obtained across the subjects considered in this study. Its origins appear to be multifactorial. In the first place, considerable differences in the levels of experience that the study participants had in undergoing BCI experiments, reflected in a varying degree of their BCI skills, should be emphasized. It is also related to differing levels of subjects' motivation and immersion in the BCI training paradigm observed before and after the recording sessions. Finally, it has been shown that the clarity and intensity of the MI induced ERD/ERS manifestations in EEG is highly subject-dependent [34]. Individual neurophysiological features can thus affect the user's capability to operate a BCI system. Closer inspection of the t-f power spectral distributions of the MI induced EEG trials in this study revealed that in some subjects the ERD/ERS correlates were found scarcely discernible. These observations corresponded to the level of the classification performance obtained for the given individuals (cf. Table III).

Although the feature extraction framework presented in this paper is not intrinsically causal, it can be modified to satisfy the requirements of online applicability. To this end, a moving window method with continuous classification on a sample-by-sample basis should be adopted. A current time point within a

trial should serve then as the end point of the feature extraction window. At the same time, this approach is associated with a delayed response of the BCI system to the changes in the spectral content of the ongoing EEG activity. The window size of up to 2 s, used in this work, allows for balancing the time and the frequency resolution of analysis of the spectral phenomena relevant to MI related EEG pattern discrimination. In consequence, the time delay of the causal feature extractor mentioned above is considered to be manageable and acceptable in online BCI operation. It should be emphasized that the proposed transformation enforcing causality still requires time efficient implementation of the spectral techniques. Sample-by-sample classification in online mode within the framework discussed here implies that the spectral quantification of an EEG trial within a moving window has to be performed faster than the sampling frequency, f_s . Preliminary simulations conducted in the ISRC laboratory setup (with Pentium IV 3 GHz, 512 MB RAM) demonstrated the feasibility of applying the spectral methods, including quadratic energy distributions with their highest computational complexity, in online EEG-based BCI systems with real-time constraints ($f_s \leq 128$ Hz for moving windows of equal or less than 2 s duration, which corresponds to the maximum of 256 data points processed within 1/128 s).

V. CONCLUSION AND FUTURE WORK

This paper has investigated various approaches to quantifying the relevant spectral content of the EEG utilized for MI classification. The separability of the EEG feature sets representing entire BCI trials has been evaluated offline. Special attention has been paid to session-to-session generalization properties of the classifiers and the feature sets.

A detailed ranking of the most effective spectral methods in terms of the resultant CA rates is rather subject-specific. There are a few cases of techniques achieving good performance in one subject and poor performance in the other one. Similarly, high CA rates in one session do not necessarily correspond with satisfactory results in the other session. Close observation of the trends across all eleven subjects has demonstrated the superior robustness of PSD approaches consistent with all the classifiers used in one- and two-session classification tasks. The most inferior session-to-session stability was produced by the Yule_{AR} approach.

With regard to classification methods, the study has revealed the superiority of SVMs with Gaussian kernel (SVM_{Gauss}) in Session I experiments for the majority of the feature types. Clearly poorer performance is observed only for the Yule_{AR} approach. In session-to-session experiments, the selection of an optimal classifier that robustly maintains good performance from one session to the other one is more subject and feature type dependent. Generally, the consistent effectiveness of linear approaches investigated in this work, L2-SVM_{lin}, LS-SVM_{lin}, and RFD in particular, is emphasized.

Although the concept of discrete classification performed at the end of each trial is less practical in real-world BCI applications than a continuous approach, this thorough study still gives a valuable insight into the separability of MI induced EEG patterns, in the same spirit as in [9], [17], and [18]. It also offers some informed guidance as to which techniques are more suit-

able for BCI. The combination of different EEG datasets with a range of motor imagery dynamics from beginning to end of trial obtained during MI-based BCI experiments, undertaken by subjects with varying levels of prior experience and capabilities in using EEG-based communication devices, enabled the production of more general conclusions at the cost of relatively lower average performance and higher intersubject variance. The extensive complementary online analysis, within the framework of causal feature extraction with a moving window approach and continuous classification, is intended for further work.

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